

NEXORA

Next-Generation Emotional Resilience & Distress-Monitoring Assistant for Victims in Legal & Rehabilitation Proceedings.

Role: Human-in-the-Loop Clinical AI Platform

AI Model: Google Gemini 1.5/2.0 Flash

UI: React 19 + Vite + Tailwind CSS

Target: Legal, Police & Health Casework

1. Problem Statement & The NEXORA Solution

The Critical Problem

Victims of crime and trauma undergo protracted legal trials lasting months or years. Caseworkers handle 50+ cases simultaneously, resulting in infrequent check-ins. Emotional distress, anxiety, and suicidal ideation are usually detected **reactively after an emergency occurs**.

The NEXORA Innovation

NEXORA transforms victim support from reactive crisis control into **proactive prevention**. It provides continuous, empathetic AI check-ins, calculates clinical distress metrics, and alerts counsellors immediately when trauma thresholds are breached.

2. Role-Based Access Control (RBAC) & Portals

Portal Role	Core Functionality	Impact & Clinical Value
VICTIM PORTAL	Empathetic AI Companion chat, conversational stress analysis (High/Medium/Low), integrated 'Consult a Doctor' portal for instant specialist tele-consultations, and milestone tracking.	Provides safe, judgment-free expression with no clinical stigma, available 24/7 in native languages.
COUNSELLOR PORTAL	Clinical triage dashboard, active distress telemetry (0-100), predictive trend warnings, and clinical notes management.	Enables counsellors to prioritize high-risk victims immediately and conduct targeted interventions.
AUTHORITY / ADMIN	Aggregated caseload analytics, resolution rates, caseworker workload distribution, and judicial phase tracking.	Gives administrative leaders complete institutional oversight and optimal resource deployment.



3. End-to-End System Workflow

STEP 1

Victim Interaction

Daily check-in / freeform
chat with Gemini AI
Companion

STEP 2

Distress Scoring

Algorithm scores PHQ-9 /
GAD-7 metrics (0–100)

STEP 3

Predictive Trigger

Spike detected (>75) →
automated high-priority
alert

STEP 4

Human Intervention

Counsellor reviews XAI
insights and schedules
outreach



4. 10 Core Architectural & AI Capabilities

1 Dynamic Distress & Stress Monitor

Continuous 0–100% scoring calculated from conversational sentiment, linguistic distress markers, and check-in surveys.

2 Predictive Distress Engine

Trajectory forecasting using historical trend curves to identify crisis points before they occur.

3 Multimodal Analysis

Multi-variable assessment synthesizing user sentiment, response frequency, and semantic markers.

4 Adaptive Clinical Check-ins

Grounded in clinical psychology standards (PHQ-9 for depression, GAD-7 for generalized anxiety).

5 AI Companion & Concluding Stress Assessment

Google Gemini-powered companion that evaluates conversational distress and provides an instant 0–100% stress breakdown with emotional tone telemetry on session conclusion.

6 Crisis & Threat Detection

Instant detection of self-harm or critical triggers with automatic emergency hotline injection (Tele-MANAS, 1091).

7 Smart Alert Escalation Matrix

Automated tiering (Low, Moderate, High, Critical) ensuring prompt notification to caseworkers.

8 Human-in-the-Loop Interventions

AI recommends; human experts decide. Caseworkers log notes, resolve alerts, and schedule sessions.

9 Victim Case Journey Tracker

Visual legal & recovery milestone tracker (FIR, Court Hearings, Therapy, Rehabilitation).

10 Explainable AI & Privacy (XAI)

Transparent rationales detailing why scores altered, maintaining ethical and clinical compliance.



5. Full Technology Stack Breakdown

Layer	Technologies Used	Implementation Details
Frontend Application	React 19, TypeScript, Vite, Tailwind CSS	High-performance Single Page Application (SPA), responsive medical-grade design, Dark/Light mode theme system.
Visualizations & UI	Recharts, Lucide Icons, Canvas-Confetti	Predictive distress area/line charts, triage badges, milestone celebration micro-interactions.
Artificial Intelligence	Google Gemini API (Flash models), Prompt Engineering	Empathetic conversation engine, sentiment analysis, clinical distress keyword extraction, safety guardrails.
Clinical Scoring Engine	Custom TypeScript Algorithm (`aiService.ts`)	PHQ-9 and GAD-7 mathematical weighting, crisis keyword triggers, threshold evaluation.
State & Data Layer	React Context API (`useStore`), LocalStorage	Role-based authentication, persistent session management, reactive state updates across modules.
Version Control & Tooling	Git, GitHub, ESLint, npm	Modular feature branching, strict linting, clean component separation.



6. 60-Second Elevator Pitch for Judges

"Good morning judges. In our legal and rehabilitation systems, victims suffer in silence during long legal battles while counsellors are overwhelmed with 50+ cases. NEXORA bridges this gap with a Human-in-the-Loop AI platform. On the victim side, a 24/7 empathetic Gemini-powered companion provides safe check-ins. On the caseworker side, an intelligent clinical triage dashboard tracks a Dynamic Distress Score and predicts emotional downturns before a crisis occurs. With strict safety guardrails and explainable AI, NEXORA ensures that no victim falls through the cracks."